Quick quiz: Lecture 5

Scientific Computing 2026 5 minutes 5 points

Name: _____ Date: _____

Choose **one answer per question**. Each question is worth one point. No calculator is needed.

1. Method of moments

Independent waiting times follow an exponential model with unknown rate λ. Their observed mean is 4 minutes. Using $\mathbb{E}_\lambda[T] = 1/\lambda$, what is the method-of-moments estimate of λ, in requests per minute?

A. 4 **B.** 1/16 **C.** 1/4 **D.** 2

2. Gaussian maximum likelihood

Four independent Gaussian measurements have observed mean $\bar{x} = 5$ and $\sum_{i=1}^{4}(x_i - \bar{x})^2 = 12$. Both the population mean and variance are unknown. What is the maximum likelihood estimate of the **variance**?

A. 3 **B.** 4 **C.** $\sqrt{3}$ **D.** 12

3. Bias and mean squared error

For the same scalar parameter, estimator A has bias 0 and variance 0.09. Estimator B has bias 0.1 and variance 0.04. Which estimator has smaller mean squared error (MSE), and why?

A. A, because its MSE is 0.09, whereas B's is 0.14.

B. A, because an unbiased estimator always has smaller MSE.

C. B, because its MSE is 0.04.

D. B, because its MSE is 0.05.

4. A probability model gives a training loss

Assume conditionally independent responses $Y_i \mid x_i \sim \mathcal{N}(f_w(x_i), \sigma^2)$, with a known, shared $\sigma^2 > 0$. Maximizing the conditional likelihood over w is equivalent to minimizing which expression?

A. $\sum_i |y_i - f_w(x_i)|$.

B. $\sum_i (y_i - f_w(x_i))^2$.

C. $\sum_i (y_i - f_w(x_i))$.

D. $\sum_i f_w(x_i)^2$.

5. MAP and a Gaussian prior

For an unknown mean μ, the likelihood favors $\mu = 2$, while a Gaussian prior favors $\mu = 0$. The negative log-posterior, up to a constant, is

$$J(\mu) = \frac{1}{2}(\mu - 2)^2 + \frac{1}{2}\mu^2.$$

What is the maximum a posteriori (MAP) estimate?

A. 1 **B.** 0 **C.** 2 **D.** 4